

CapScan

Introduction

CapScan is an innovative software project designed to provide comprehensive analysis and testing of storage devices. It verifies storage capacity, checks data integrity, evaluates read/write speeds, and ensures long-term reliability. CapScan addresses the issue of counterfeit or faulty devices by equipping users with tools to accurately assess and trust their storage devices. This helps mitigate risks like file corruption and data loss.

Key Features

Capacity Scanning

CapScan measures the actual usable capacity of a storage device and compares it with the manufacturer's claimed size. This enables users to immediately detect counterfeit or faulty devices, ensuring that they do not rely on misleading specifications.

Integrity Verification

The software performs detailed integrity tests to detect corruption, incomplete writes, or data loss. By writing and reading back sample data, CapScan ensures that the stored files are safe and free from silent corruption.

Speed Benchmarking

CapScan includes advanced benchmarking tools to test both read and write speeds under multiple conditions. This allows users to evaluate performance for everyday use cases such as file transfer, video editing, or system backups.

Reliability Assessment

Beyond one-time testing, CapScan can analyze long-term device performance, monitoring error rates and performance stability over time. This helps predict failures and increases user confidence in storage reliability.

Technical Importance

CapScan holds significant value in both personal and professional contexts.

Personal Users can confirm the authenticity of USB drives, memory cards, or external HDDs before storing photos, documents, or backups.

Professionals in IT, media production, and enterprise environments can use CapScan as a diagnostic tool to ensure that high-performance SSDs, RAID setups, and backup drives are trustworthy.

Forensic and Data Recovery Experts gain a dependable solution for verifying evidence storage devices and preventing further corruption.

By combining capacity testing, integrity checks, and performance monitoring in one platform, CapScan goes beyond traditional diagnostic utilities.

Use Cases

1. Detecting fake memory cards that falsely claim higher storage capacity.
2. Ensuring reliability of external drives used for critical backups.
3. Benchmarking SSDs for high-performance applications such as gaming, video editing, or cloud servers.
4. Verifying storage integrity in forensic investigations and recovery labs.
5. Providing peace of mind to students, professionals, and businesses that rely on digital storage daily.